DawaLink

A REAL PHARMACY QUESTION, MADE EASIER · OCTOBER 2026

“3afak wach kayn Doliprane 1g f Rabat?”

A customer wants a clear answer about one exact medicine pack. DawaLink helps them identify it and prepare a useful pharmacy question—in Darija, Arabic or French.

The product promise

Understand the request
Confirm the exact catalogue item
Prepare an editable enquiry

The pharmacy supplies the real stock answer.

Local-first · 9,908 imported catalogue rows · free/open-source runtime · no hosted AI or paid API

User story · from uncertainty to a useful question

THE EXPERIENCE

Customer need

As a person looking for a medicine, I want to explain what I need in my own language, so I can identify the right catalogue item and ask a pharmacy clearly without guessing availability.

1 · Explain the need

“3afak wach kayn Doliprane 1g f Rabat?”

The local model extracts intent, medicine, strength and city.

2 · Confirm the pack

DawaLink shows three real 1 G records. The customer compares form and package, then selects ID 563.

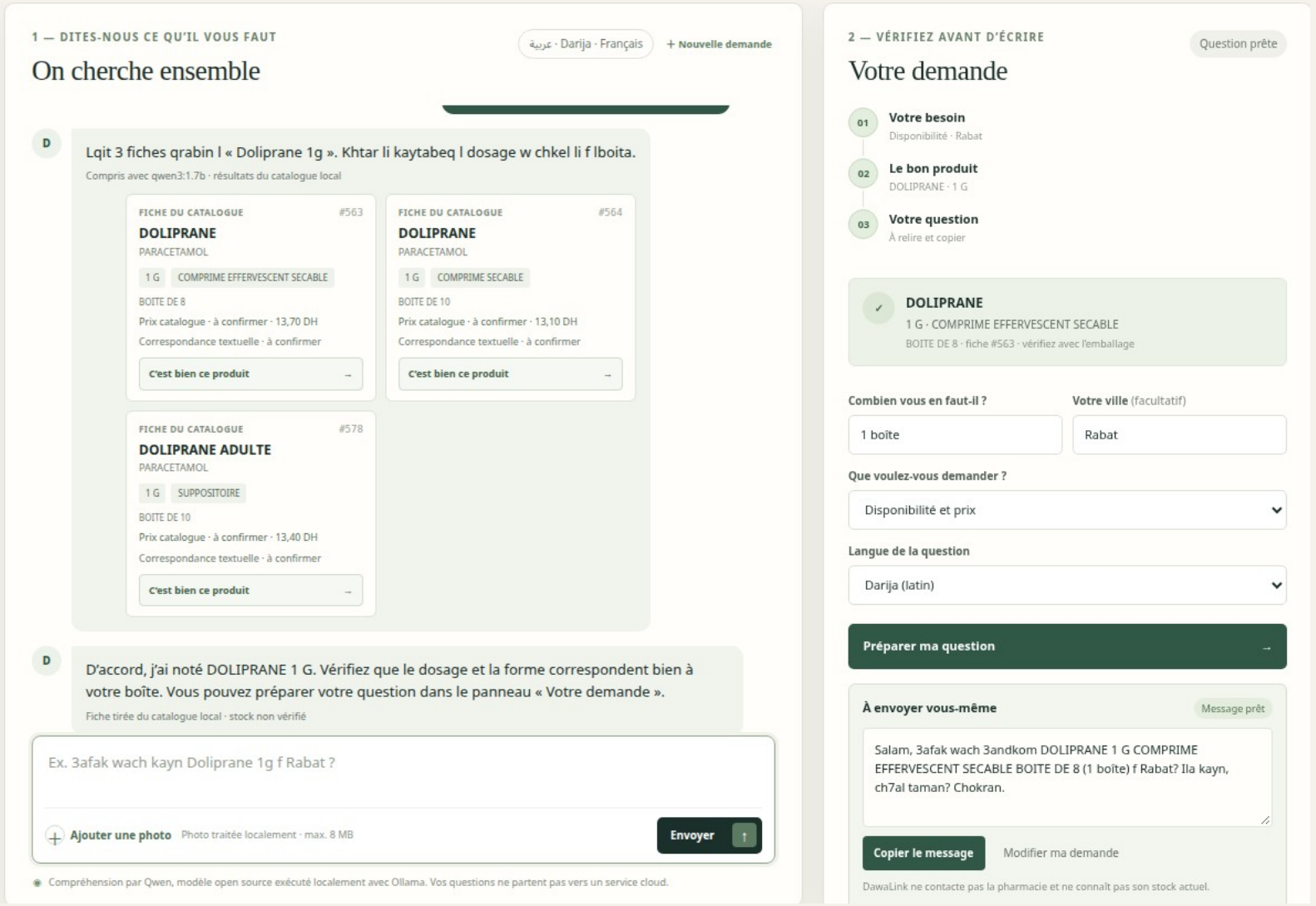
3 · Prepare the question

Review the quantity, city and Darija wording. Copy it to contact a pharmacy.

Success means an exact, customer-confirmed product and a useful enquiry. The customer decides what to send; the pharmacy provides the actual stock answer.

The customer stays in control at every step

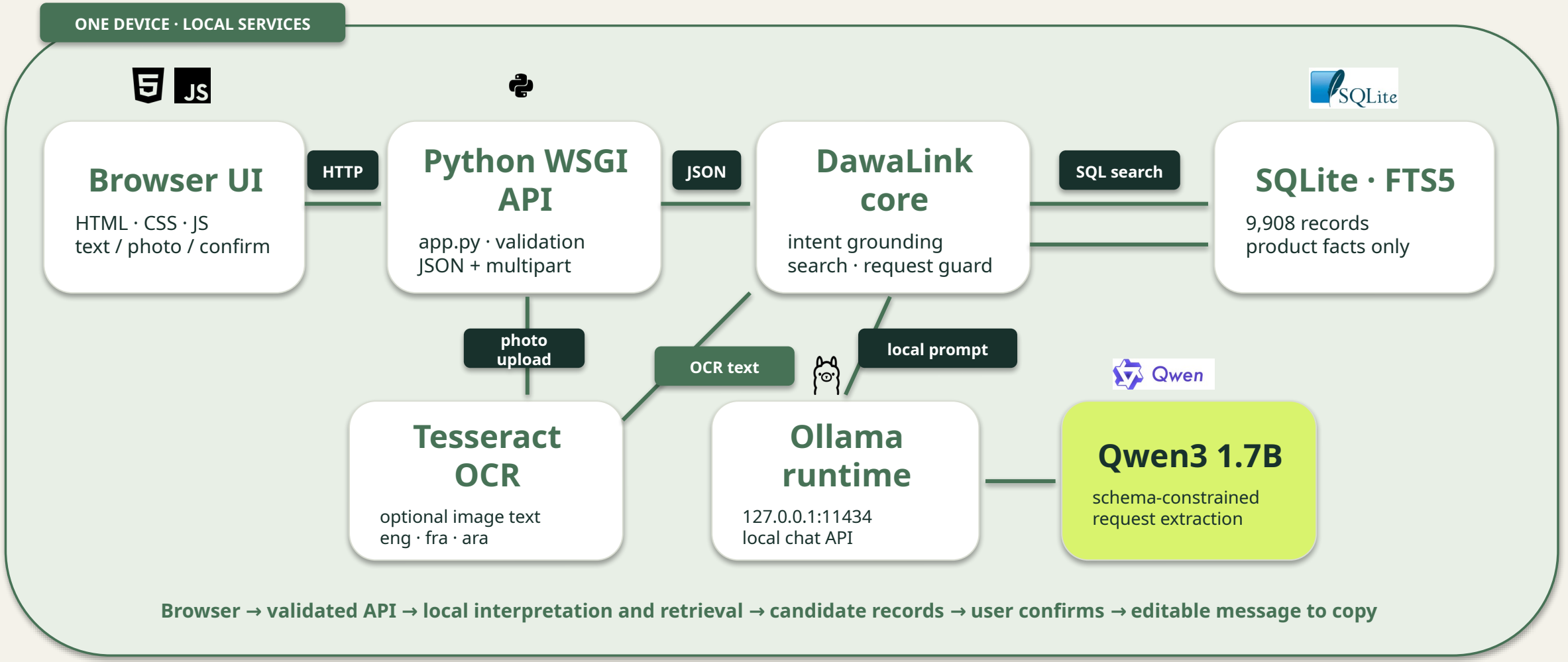
OBSERVED LIVE WORKFLOW



Captured from the running app · selected ID 563 · customer copies the message; inventory is not verified.

Technical architecture · local, grounded, inspectable

RUNTIME DESIGN



No cloud LLM · no external pharmacy directory · no live inventory · no automatic sending

The model interprets intent; the catalogue supplies facts

TRUST BOUNDARY



The model is a language interface—not a medicine database. Search results are grounded, strength conflicts are filtered, and the customer confirms the exact pack.

Real catalogue data, with its gaps kept visible

DATA PIPELINE

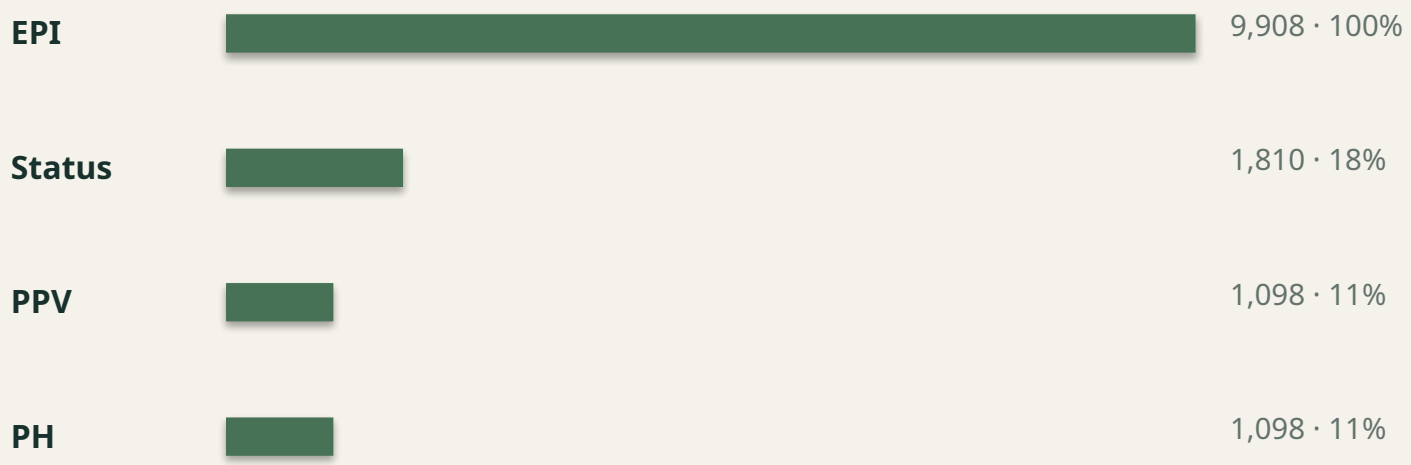
9,908 source rows

10 original columns

0 duplicate IDs

4,070 brand strings

Missing or unavailable values in supplied export



Source CSV
9,908 rows

ingest_data.py
normalize + validate

SQLite · FTS5
atomic rebuild

Provenance matters

Repository export date is unknown. Values are preserved as supplied; historical price fields are not live pharmacy prices. No missing manufacturer, stock or status facts are filled in.

Search narrows the choice; the customer makes it

CATALOGUE RETRIEVAL

Query	Observed result	What the system uses
doliprene	DOLIPRANE · 1 G	normalized + fuzzy text
AMOXICILLINE	ALFAMOX · 1 G	active ingredient
DOLIPRANE 500 MG	DOLIPRANE · 500 MG	strength agreement
DIACURE	no result	weak lookalike suppressed

Search scores are matching signals, not calibrated probabilities. Similar products remain separate choices; strength and form must be checked.

Photo identification is an optional local path

IMAGE PROCESSING

1 · Validate file type and size

2 · Preprocess locally

3 · Tesseract reads visible text

4 · Search the same SQLite catalogue

5 · Customer reviews any candidate

Evidence: one real photo spot check

MIT test image showed a foreign-market DiaCURE pack. OCR read ENFANTS / CHILDREN and missed the brand; catalogue result was safely empty.

This is a documented failure case—not a performance claim. A Moroccan product photo benchmark is still needed.

Small API, clear responsibilities

IMPLEMENTATION

Browser ↔ Python WSGI

GET /api/health
GET /api/medications?q=...
GET /api/medications/{id}
POST /api/assistant/message
GET /api/assistant/status
POST /api/recognize
POST /api/pharmacy-request
GET /api/export

SQLite storage

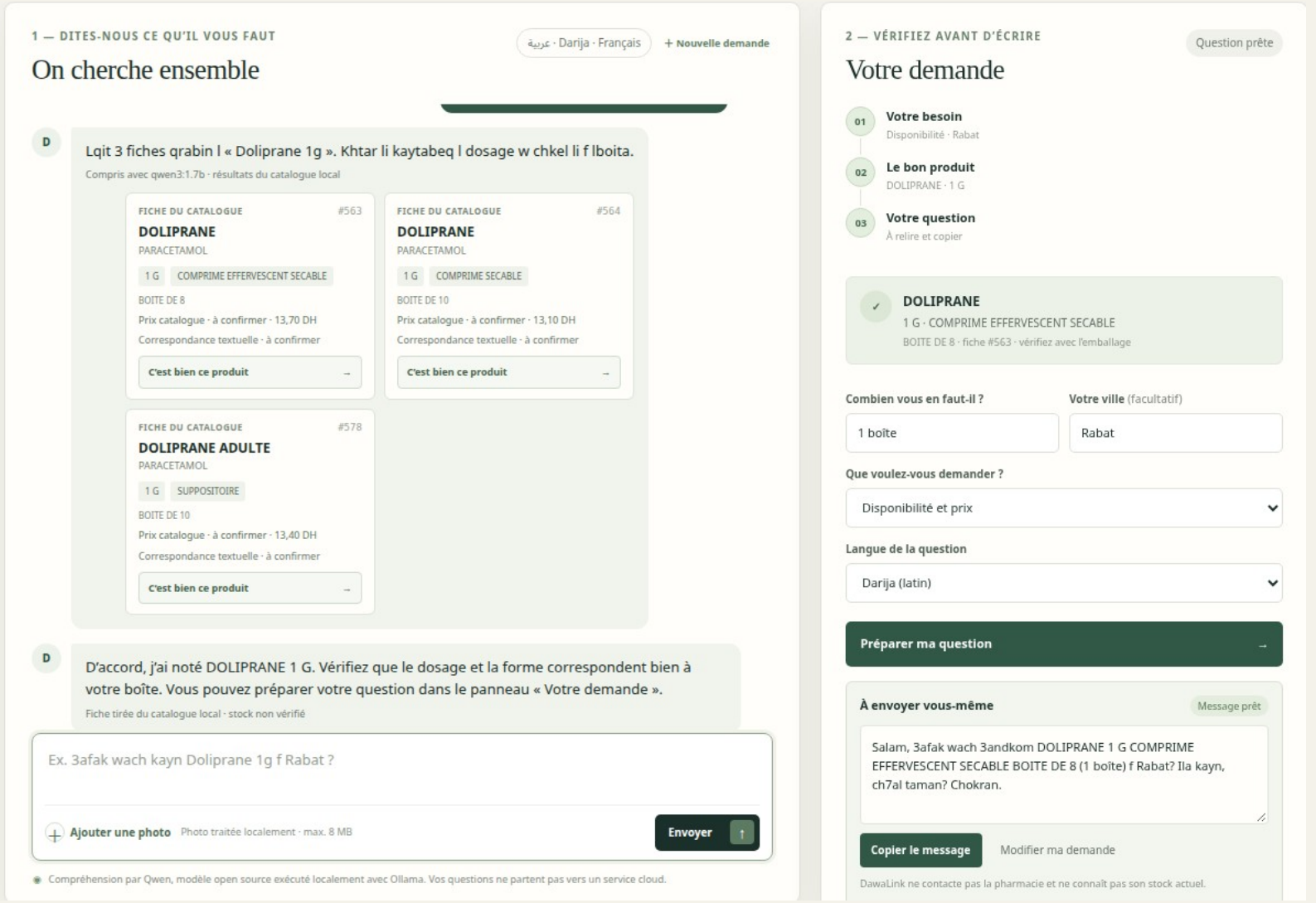
Original source columns + normalized search keys, source, import timestamp; indexed FTS5 product text.

Enquiry guard

Generation requires a valid catalogue ID. User reviews the product, quantity, city and wording, then copies the message.

A clear enquiry the customer can review and copy

PRODUCT OUTCOME



Product ID 563 · DOLIPRANE 1 G · BOITE DE 8 · 1 boîte · Rabat · captured in the running app

What has been verified—and what has not

EVALUATION

Core logic

18 earlier tests passed
6/6 catalogue probes hit top 3

LLM flow

3 live smoke prompts
Darija · Arabic · French
No accuracy benchmark

Photo evidence

1 foreign photo
brand missed
No Morocco benchmark

Scope today: conversational intent extraction, catalogue search, optional text OCR and message preparation.
Not present: diagnosis, dosing advice, live prices, stock, pharmacy directory, voice, ordering or automatic messaging.

Run the same local stack

REPRODUCIBILITY

1 · Load the model

ollama pull qwen3:1.7b

Start Ollama if needed: ollama serve

2 · Prepare and launch

python3 scripts/ingest_data.py
python3 app.py

Open http://127.0.0.1:8000

Ollama serves Qwen3 1.7B on loopback. Tesseract 5 and language packs are optional for image OCR. Model weights are downloaded separately; no paid service is required.